Appendix

Appendix A

Appendix Table A1: The timing of "post-score preference submission" reform across Chinese provinces

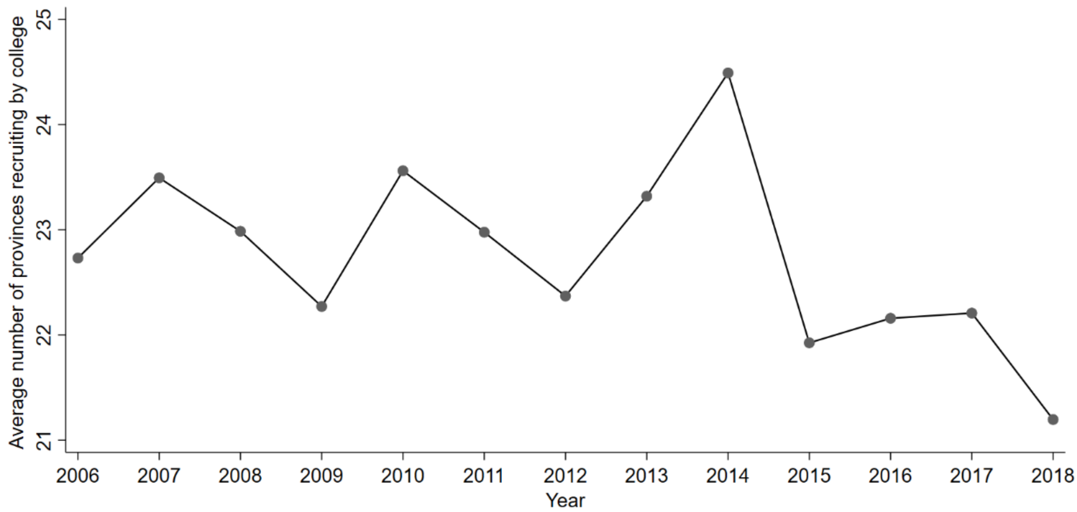
before 2005	2005	2006	2007	2008	2010
Guangxi	Fujian	Chongqing	Anhui	Gansu	Henan
Hainan	Sichuan		Jiangxi	Guangdong	Shanxi
Hebei				Guizhou	
Hubei				Jilin	
Hunan					
Jiangsu					
Neimenggu					
Ningxia					
Qinghai					
Shandong					
Xizang					
Yunnan					
Zhejiang					
2011	**2012**	**2013**	**2014**	**2015**	**2017**
Tianjin	Shanxi*	Heilongjiang	Liaoning	Beijing	Shanghai
				Xinjiang	

Note: Data prior to 2011 are drawn from Kang and Ha (2016), while data from 2011 onward are collected from provincial government websites. Shanxi*means "west of the mountains" in Chinese.

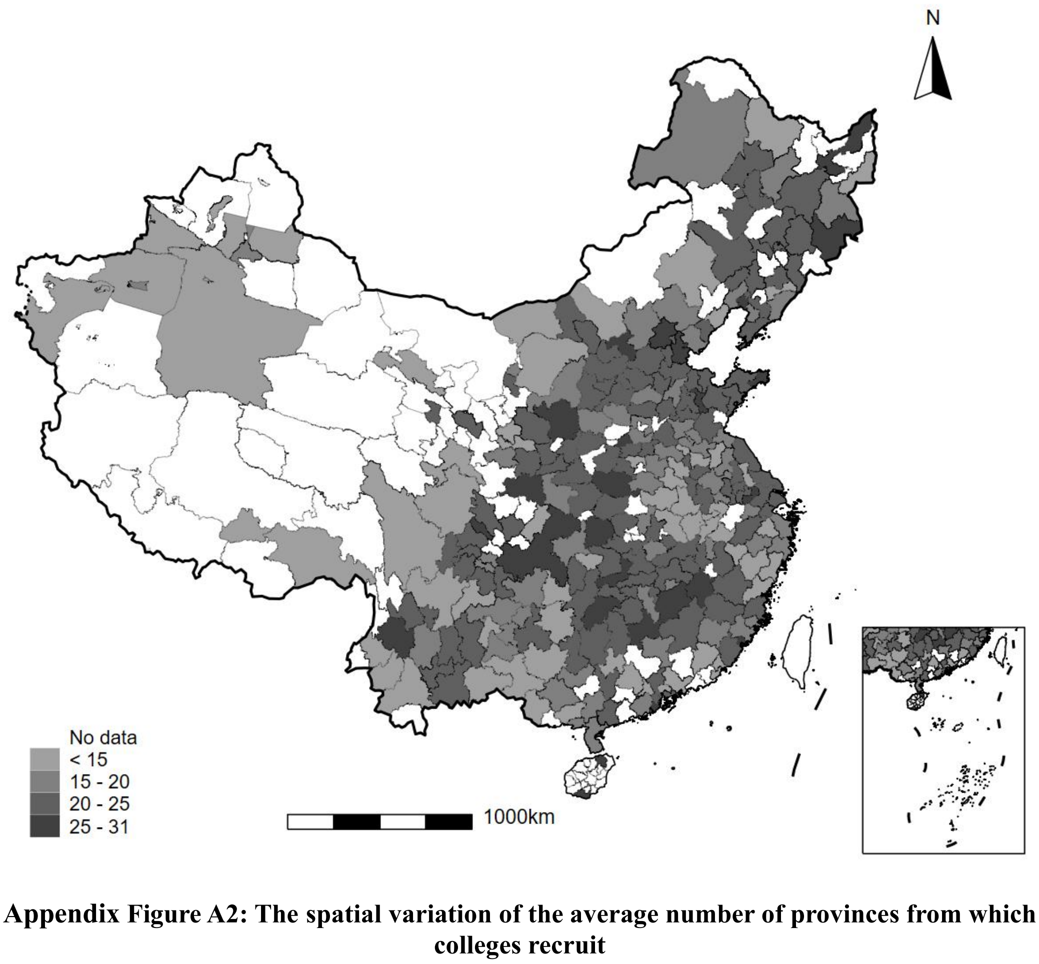
Appendix Table A2: The timing of "parallel preference" reform across Chinese provinces

2003	2005	2007	2008	2009	2010
Hunan	Jiangsu	Zhejiang	Anhui	Fujian	Chongqing
			Liaoning	Guangxi	Guangdong
			Shanghai	Guizhou	Henan
				Hainan	Shanxi
				Hebei	Tianjin
				Jiangxi	Xizang
				Jilin	
				Ningxia	
				Sichuan	
				Yunnan	
2011	**2012**	**2013**	**2014**	**2015**	**2018**
Hubei	Shanxi*	Heilongjiang	Beijing	Gansu	Qinghai
Xinjiang		Shandong			

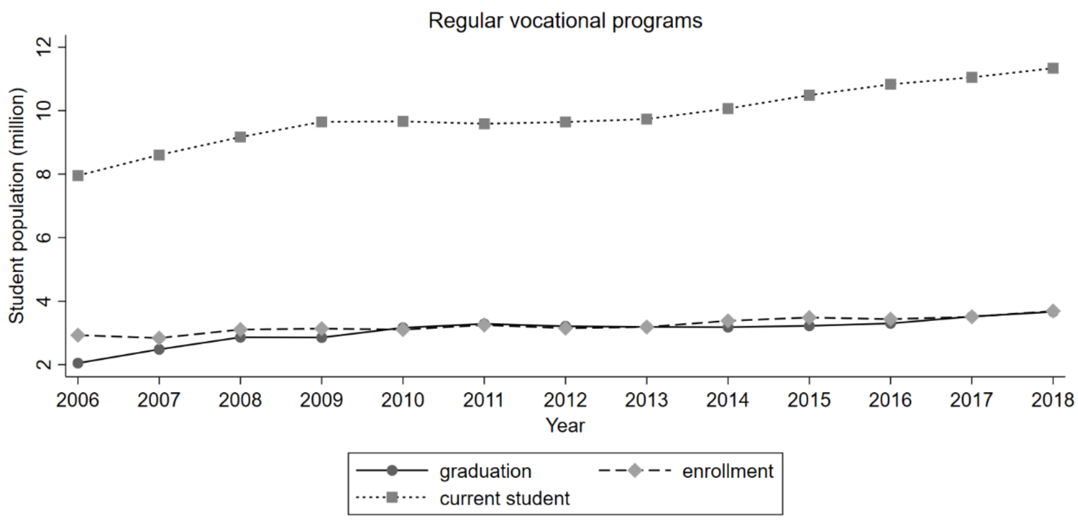
Note: The data are drawn from Sun and Liu (2025). As of 2018, Inner Mongolia was the only province that had not implemented the parallel preference mechanism. Shanxi*means "west of the mountains" in Chinese.



Appendix Figure A1: The time trend of the average number of provinces from which colleges recruit

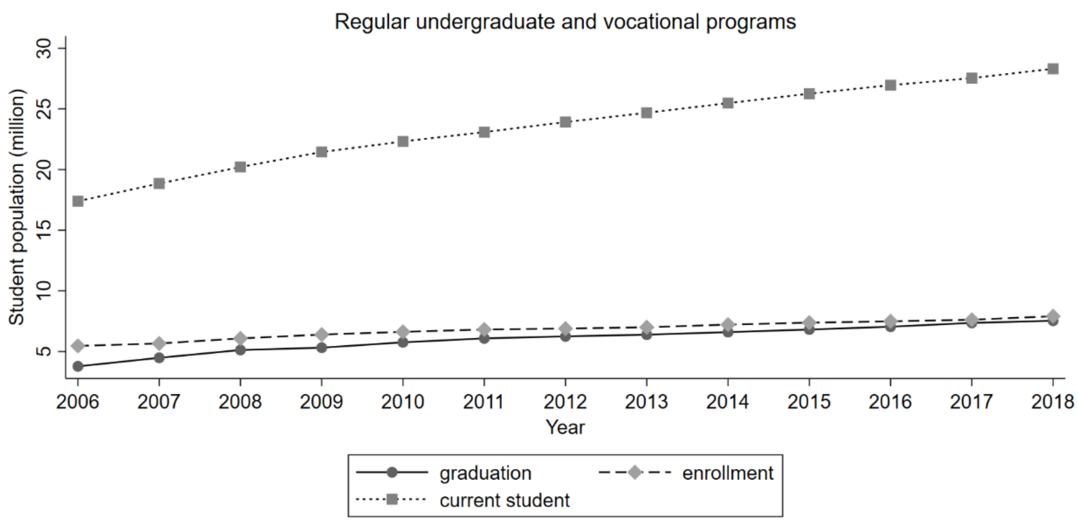


Appendix Figure A2: The spatial variation of the average number of provinces from which colleges recruit



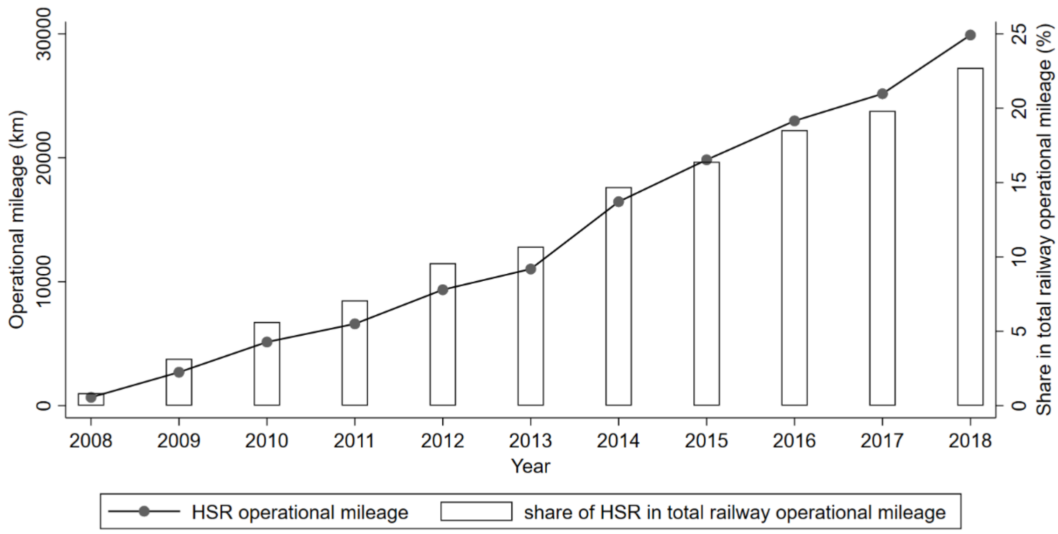
Appendix Figure A3: Student population of regular vocational programs in China

Source: *China Education Statistical Yearbook*



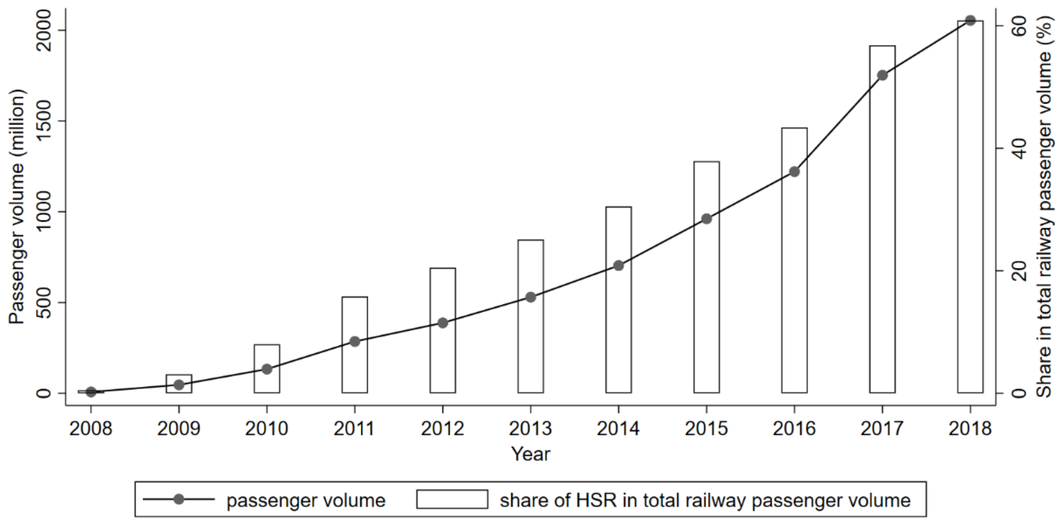
Appendix Figure A4: Student population of regular undergraduate and vocational programs in China

Source: *China Education Statistical Yearbook*



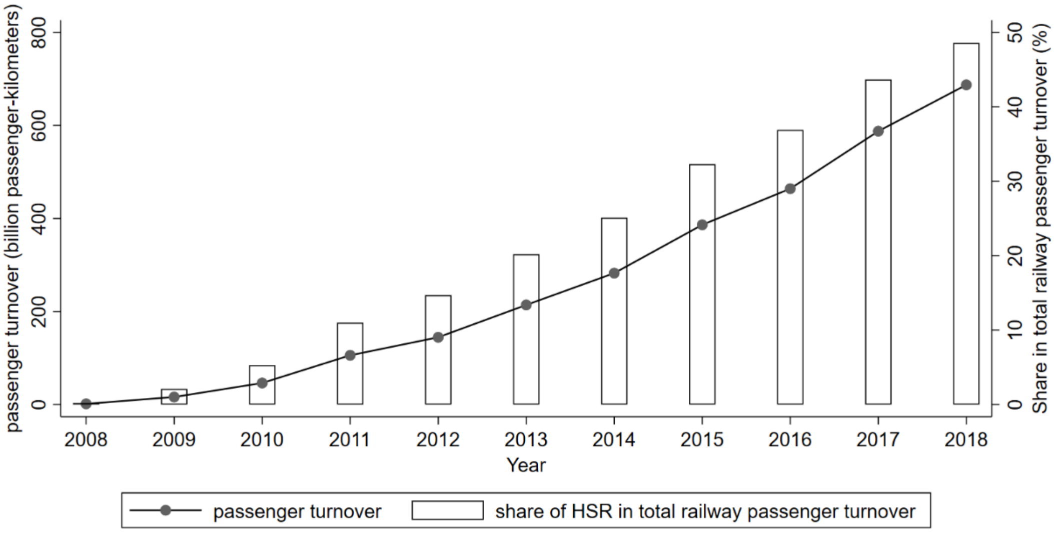
Appendix Figure A5: Total HSR operational mileage in China

Source: *China Statistical Yearbook*



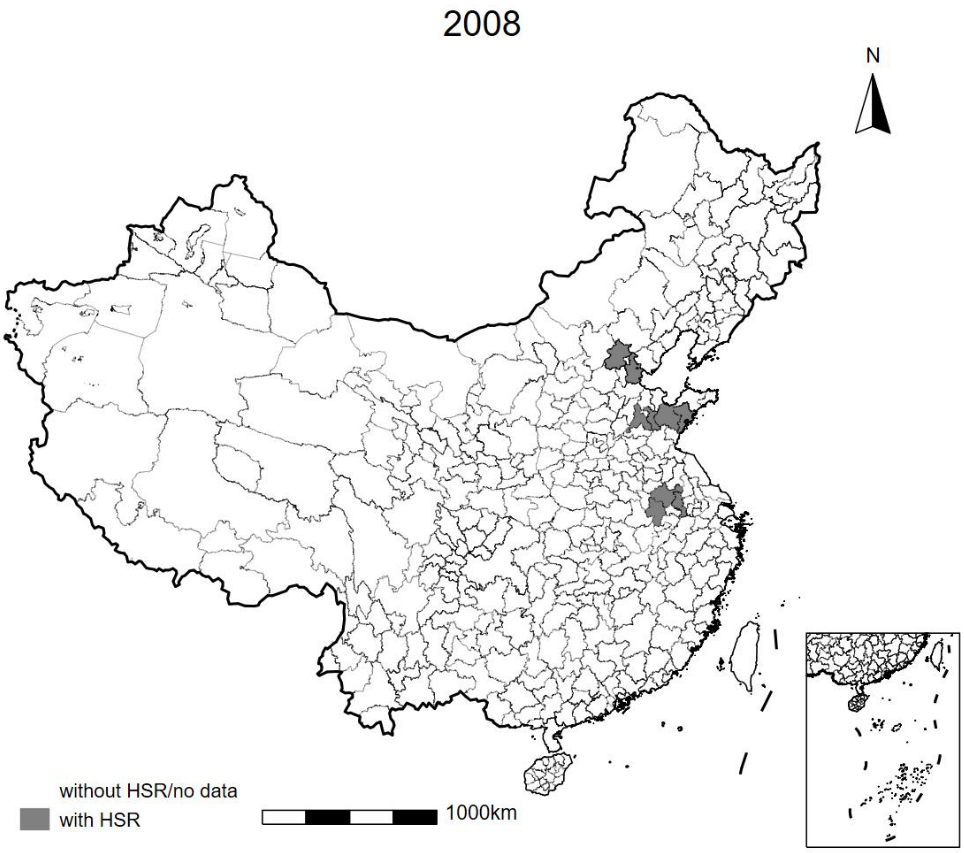
Appendix Figure A6: Total HSR passenger volume in China

Source: *China Statistical Yearbook*



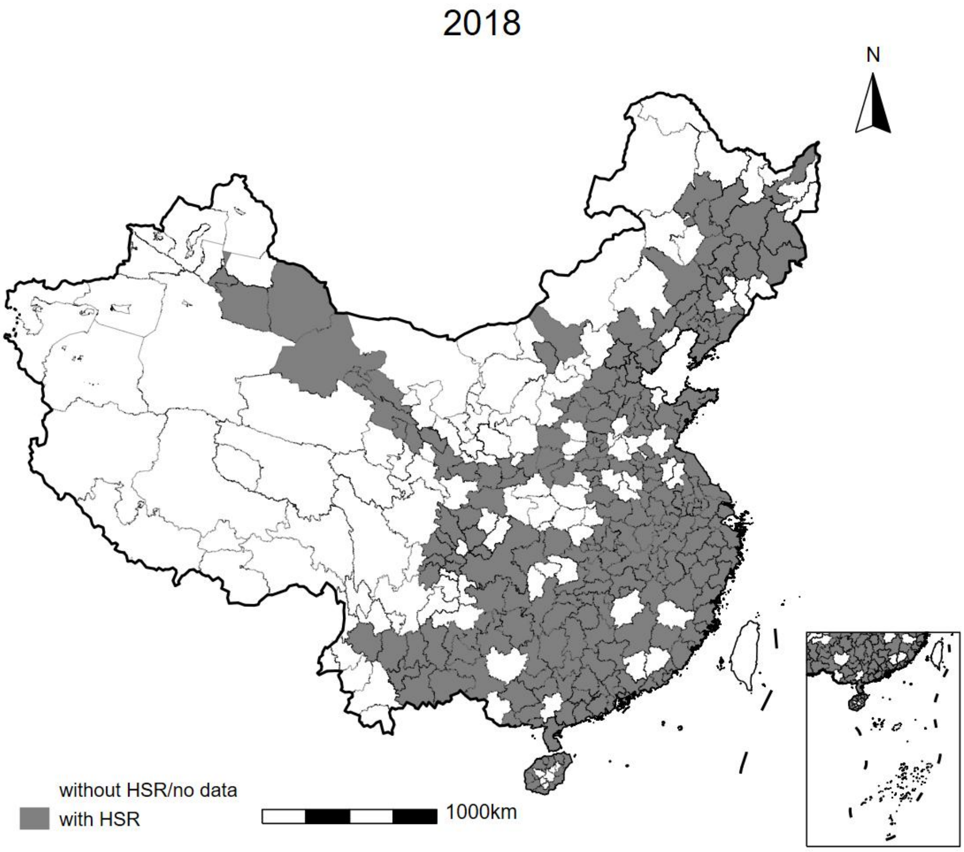
Appendix Figure A7: Total HSR passenger turnover in China

Source: *China Statistical Yearbook*



Appendix Figure A8: Distribution of HSR opening in China in 2008

Source: Chinese Research Data Services Platform (CNRDS) database



Appendix Figure A9: Distribution of HSR opening in China in 2018

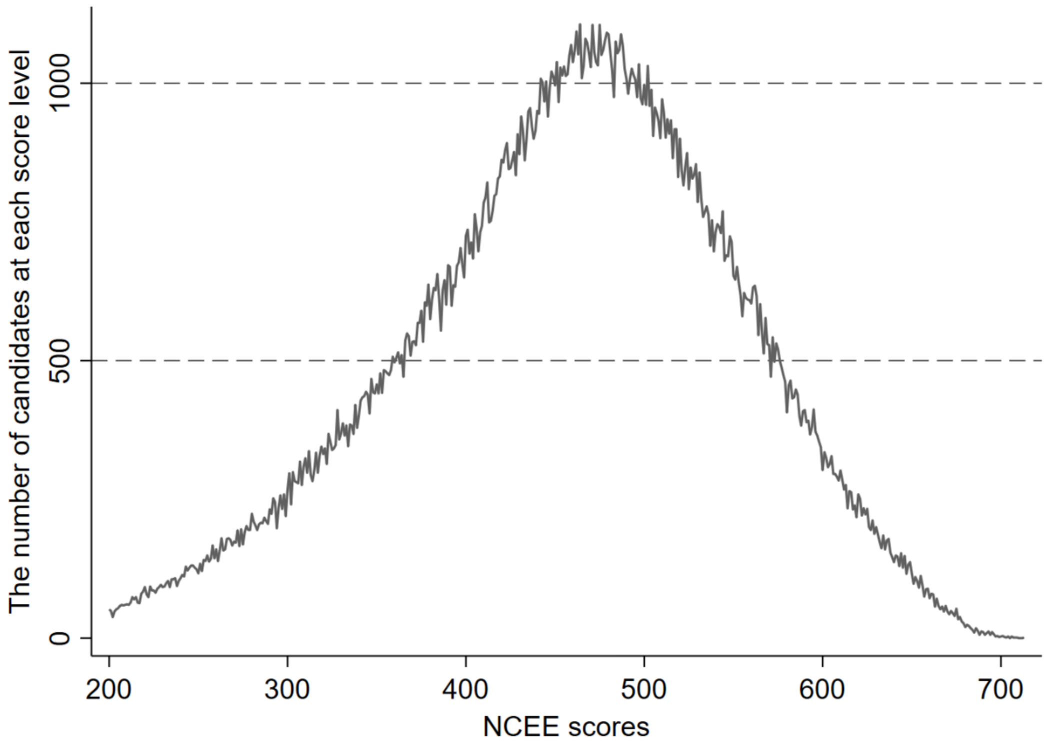
Source: Chinese Research Data Services Platform (CNRDS) database

Appendix B

Appendix Table B1: Explanation and descriptive statistics for the major variables

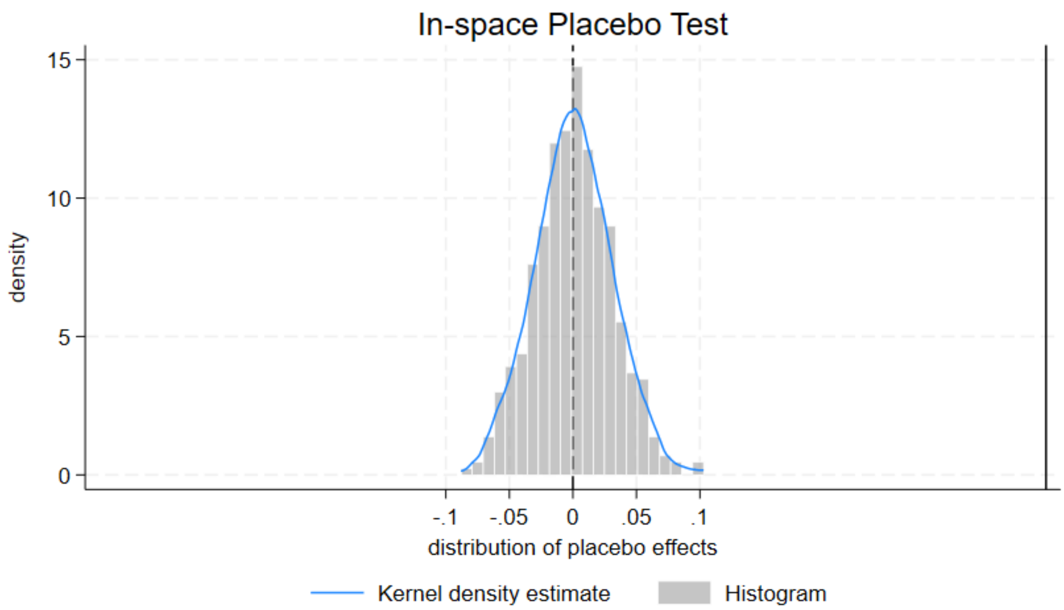
Variable	Explanation	N	Mean	Sd Dev	Min	Max
average_n	average admission scores (normalization)	416,124	40.30	21.66	0	100
hsr	A dummy variable for HSR opening	416,124	0.52	0.50	0	1
ln*gdp*	Natural logarithm of GDP per capita	416,124	16.97	1.00	13.00	19.02
popden	Population density	416,124	0.08	0.11	0.00	1.21
second	Value added share of the secondary industry	416,124	45.20	9.78	14.95	85.64
third	Value added share of the third industry	416,124	47.69	12.08	11.22	80.98
college_num	Number of colleges	416,124	36.10	26.78	1	92
ln*teacher*	Natural logarithm of teachers number in college	416,124	9.56	1.24	4.68	11.16

Appendix C



Appendix Figure C1: The number of candidates achieving each score (Anhui province, science track, 2018)

Source: https://www.gaokao.cn/



Appendix Figure C2: Placebo test results

Note: We use Stata command **didplacebo** randomly assign HSR opening to cities and run the regression of Equation (2) 500 times. Appendix Figure C3 plots the density distribution of the estimates. The vertical dashed line at 0.372 represents the baseline regression result in column (4) of Table 1.

Appendix Table C1: Robustness tests (I)

	(1)	(2)	(3)	(4)	(5)
hsr	0.0121**	1.426***	1.000***	0.235**	0.868***
	(0.00565)	(0.276)	(0.347)	(0.113)	(0.310)
Controls	√	√	√	√	√
College-admission province-track-batch FE	√	√	√	√	√
Year-admission province-track-batch FE	√	√	√	√	√
N	416,124	416,124	416,124	385,519	385,519
R-squared	0.963	0.851	0.977	0.946	0.962

Note: In Column (1), the dependent variable is the standardized average admission score at the year-admission province-subject level. In Column (2), the dependent variable is the normalized average admission score at the year-province-subject-admission-batch level. In Column (3), the dependent variable is the raw average admission score. In Column (4), the dependent variable is the normalized maximum admission score at the year-province-subject level. In Column (5), the dependent variable is the raw maximum admission score. The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table C2: Robustness tests (II)

	(1)	(2)	(3)	(4)	(5)
hsr	0.364***	0.366***	0.356***	0.355***	0.370***
	(0.109)	(0.109)	(0.114)	(0.121)	(0.110)
Controls	√	√	√	√	√
College-admission province-track-batch FE	√	√	√	√	√
Year-admission province-track-batch FE	√	√	√	√	√
N	416,676	417,958	414,530	408,913	415,853
R-squared	0.970	0.970	0.968	0.966	0.969

Note: In Column (1), we retain all observations, including those where the average admission score exceeds 750 or falls below 200. In Column (2), building upon Column (1), we further retain observations with average admission scores higher than those of Peking University or Tsinghua University. In Column (3), we narrow the range of average admission scores to [300, 700]. In Column (4), we winsorize the average admission scores at the 1st and 99th percentiles. In Column (5), we generate college identifiers directly from college names as they appear in the raw data. The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table C3: Robustness tests (III)

	(1)	(2)	(3)	(4)	(5)	(6)
hsr	0.332***	0.389***	0.360***	0.372***	0.303***	0.338***
	(0.112)	(0.112)	(0.114)	(0.106)	(0.113)	(0.114)
Controls		√	√	√	√	√
Lagged Controls	√	√				
College-admission province-track-batch FE	√	√	√	√	√	√
Year-admission province-track-batch FE	√	√	√	√	√	√
N	416,556	415,636	416,556	416,124	350,853	376,389
R-squared	0.969	0.969	0.969	0.969	0.973	0.965

Note: In Column (1), we replace the contemporaneous control variables with their one-year lagged counterparts. In Column (2), we include both contemporaneous and lagged control variables. In Column (3), we instead define HSR openings at the annual level and lag the variable by one year. In Column (4), we apply clustering robust standard errors at the college's provincial level. In column (5), we restrict the sample period to 2008–2018. In Column (6), we exclude Beijing and Shanghai from the sample. The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table C4: Robustness tests (IV)

	(1)	(2)	(3)	(4)	(5)
hsr	0.438***	0.387***	0.182**	0.328**	0.138*
	(0.122)	(0.105)	(0.0874)	(0.132)	(0.0708)
Controls	√	√	√	√	√
College-admission province-track-batch FE	√	√	√	√	√
Year-admission province-track-batch FE	√	√	√	√	√
City-year trend			√		√
College province-year-admission province-track-batch FE				√	√
N	414875	416124	416124	406993	405800
R-squared	0.969	0.969	0.97	0.975	0.976

Note: In Column (1), we control for the logarithm of city-level railway passenger volume, civil aviation passenger volume, and road passenger volume. In Column (2), we further include city-level annual meteorological variables, such as average wind speed, precipitation, temperature, atmospheric pressure, sunshine duration, and relative humidity. In Column (3), we add city-specific linear time trends to control for all city-level time-varying linear characteristics. In Column (4), we use college province-year-admission province-subject-batch fixed effects. In Column (5), we simultaneously include all additional controls and fixed effects used in Columns (1) through (4). The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

We restrict the sample to provinces whose admissions policies remained unchanged during the study period (2006–2018). For the "post-score preference submission" reform, as shown in Appendix Table A1, we limit the sample to the provinces highlighted in red, which consistently adopted the "post-score preference submission" policy throughout the study period. Column (1) of Appendix Table 5 in the revised manuscript reports the corresponding regression results. As shown, when the sample is restricted to provinces that continuously implemented the "post-score preference submission" policy during the study period, the core effect remains significantly positive. Similarly, for the "parallel preference" reform, as shown in Appendix Table A2, we restrict the sample to the provinces highlighted in red, which consistently adopted the "parallel preference" policy throughout the study period. Column (2) of Appendix Table 5 in the revised manuscript reports the regression results. The results indicate that the main conclusions of the paper remain robust. It should be noted, however, that this sample restriction substantially reduces representativeness. In particular, Column (2) includes only two admissions provinces. Therefore, the results in Column (2) of Appendix Table 5 in the revised manuscript should be interpreted as suggestive evidence only.

Appendix Table C5: Robustness tests (V)

	(1)	(2)	(3)	(4)	(5)	(6)
hsr	0.376***	0.692***	0.343**	0.382**	0.357**	0.400***
	(0.122)	(0.200)	(0.155)	(0.156)	(0.155)	(0.154)
Controls	√	√	√	√	√	√
College-admission province-track-batch FE	√	√	√	√	√	√
Year-admission province-track-batch FE	√	√	√	√	√	√
N	201,632	20,353	416,051	416,124	415,993	415,750
R-squared	0.967	0.965	0.973	0.974	0.973	0.974

Note: In Column (1), as shown in Appendix Table A1, we limit the sample to the provinces highlighted in red, which consistently adopted the "post-score preference submission" policy throughout the study period. In Column (2), as shown in Appendix Table A2, we limit the sample to the provinces highlighted in red, which consistently adopted the "parallel preference" policy throughout the study period. Columns (3) and (4) report weighted regressions using city-level college student numbers and teacher numbers, respectively. Columns (5) and (6) report weighted regressions using the lagged values of city-level college student numbers and teacher numbers, respectively. The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

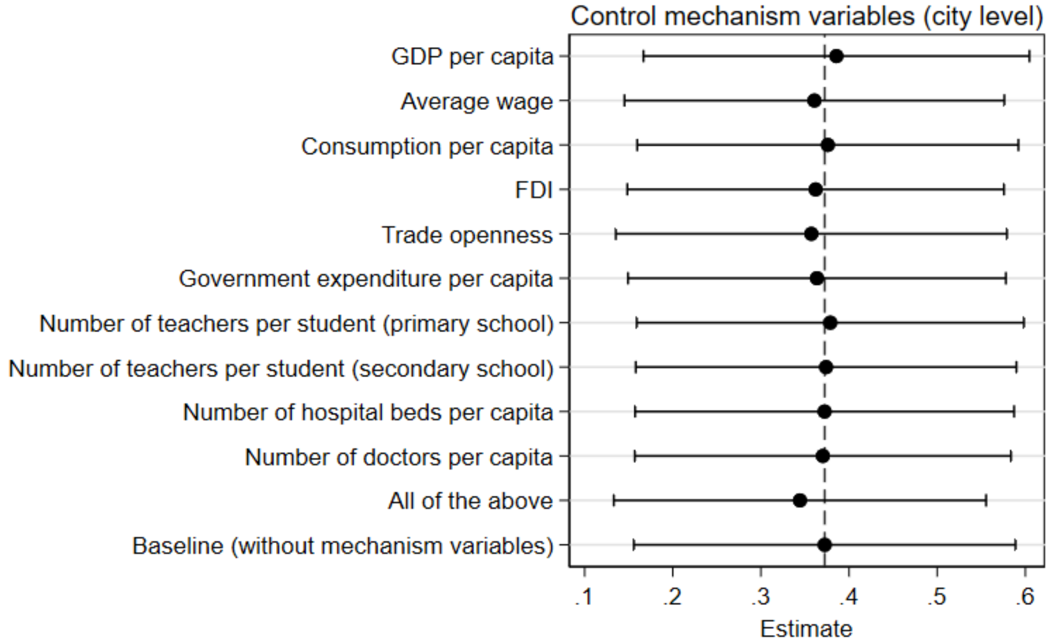
Appendix D

The first potential mechanism through which HSR may enhance college admissions quality is by promoting local economic development. Existing studies have shown that HSR stimulates urban economic growth (e.g., Ke et al., 2017; Ahlfeldt and Feddersen, 2018) and creates employment opportunities (e.g., Lin, 2017; Dong, 2018). In addition, local governments benefit from economic growth and are able to provide better public services. It is thus reasonable to expect that candidates may prefer cities with higher levels of economic development. To test this mechanism, we construct a set of city-level indicators that may influence candidates' preferences, covering income, consumption, international economic activity, and public services. Specifically, these indicators include: GDP per capita, average wage, consumption per capita, inflows of foreign direct investment (FDI), trade openness (measured as the ratio of total imports and exports to GDP), government expenditure per capita, number of teachers per student (primary school), number of teachers per student (secondary school), number of hospital beds per capita, and number of doctors per capita. Following the methodology of Chor et al. (2021) and Dippel et al. (2022), and without altering the baseline regression specification, we include these potential mechanism variables as additional controls. This allows us to examine whether HSR improves college admissions quality through its impact on local economic development.

Appendix Figure D1 reports the key estimated coefficients after sequentially including mechanism variables in the baseline regression. We add these mechanism variables one at a time and then include all of them simultaneously in the second-to-last row. The baseline regression results without any mechanism variables are presented in the last row and indicated by a vertical dashed line. The results in Appendix Figure D1 show that whether individual mechanism variables or all are included together, the core estimated coefficient remains largely unchanged compared to the baseline.

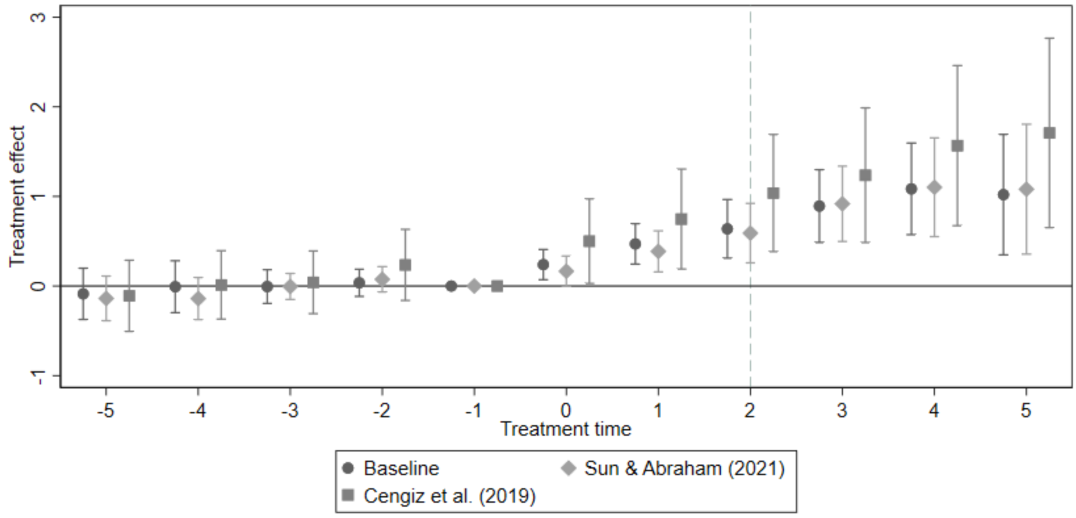
Two potential concerns may be raised. First, city-level mechanism variables could be endogenous to HSR openings. Second, using contemporaneous indicators may fail to capture the lagged effects of HSR on urban economic outcomes. To address these concerns, we conduct two additional analyses. First, we perform an event-study analysis similar to Figure 4, additionally controlling for all city-level mechanism variables. The results, reported in Appendix Figure D2, show that even after controlling for all city-level mechanisms, there is no evidence of significant pre-trends. Second, we regress city-level mechanism variables on HSR openings. Specifically, we consider both the contemporaneous impact of HSR openings and their lagged effects one, two, and three periods after opening. The results, presented in Appendix Figure D3, indicate that the vast majority of estimated coefficients are small and statistically insignificant.

Taken together, in the context of this study, the contribution of the mechanism whereby HSR affects college admissions quality through local economic development appears to be limited. It is important to clarify that this finding does not imply that HSR has no effect on city economic development, nor that candidates are unaffected by economic conditions. Addressing these two issues requires more in-depth and rigorous empirical analysis. This is beyond the scope of the present study and is left for future research. Moreover, we cannot control for all dimensions of city economic development and have only selected representative indicators. This does not preclude the possibility that HSR enhances college admissions quality through other unobserved aspects of economic development. The correct interpretation is that the potential mechanism of HSR promoting local economic development—as proxied by the selected indicators—does not explain our main finding that HSR improves college admissions quality.



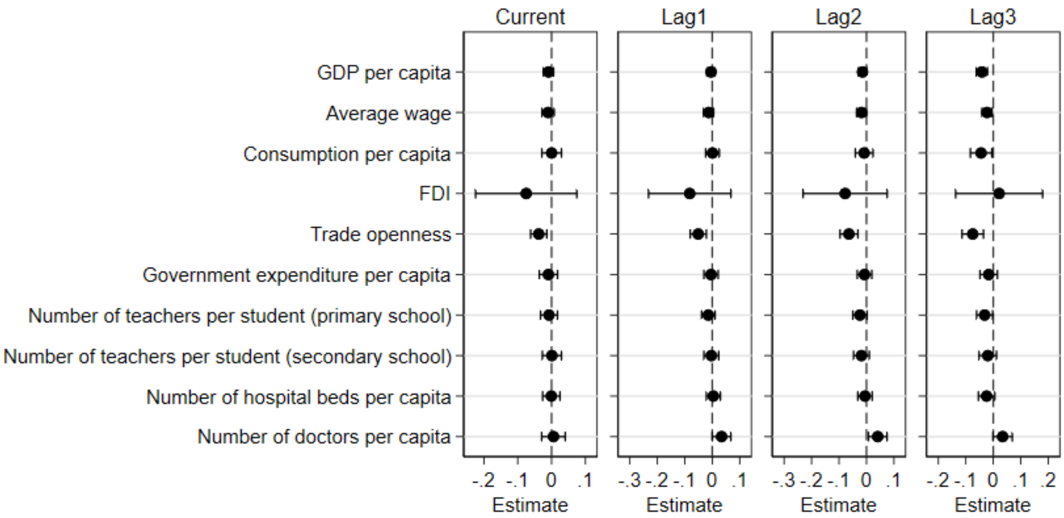
Appendix Figure D1: Potential mechanisms (HSR promotes local economic development)

Note: Appendix Figure D1 shows the estimated coefficients and 95% confidence intervals. All mechanism variables are transformed using natural logarithms, except for trade openness.



Appendix Figure D2: Event study results controlling for all city-level mechanism variables

Note: Appendix Figure D2 shows the estimated coefficients β_k and 95% confidence intervals for Equation (3) and the event study results using the methods provided by Sun & Abraham (2021) and Cengiz et al. (2019).



Appendix Figure D3: The impact of HSR openings on city-level mechanism variables

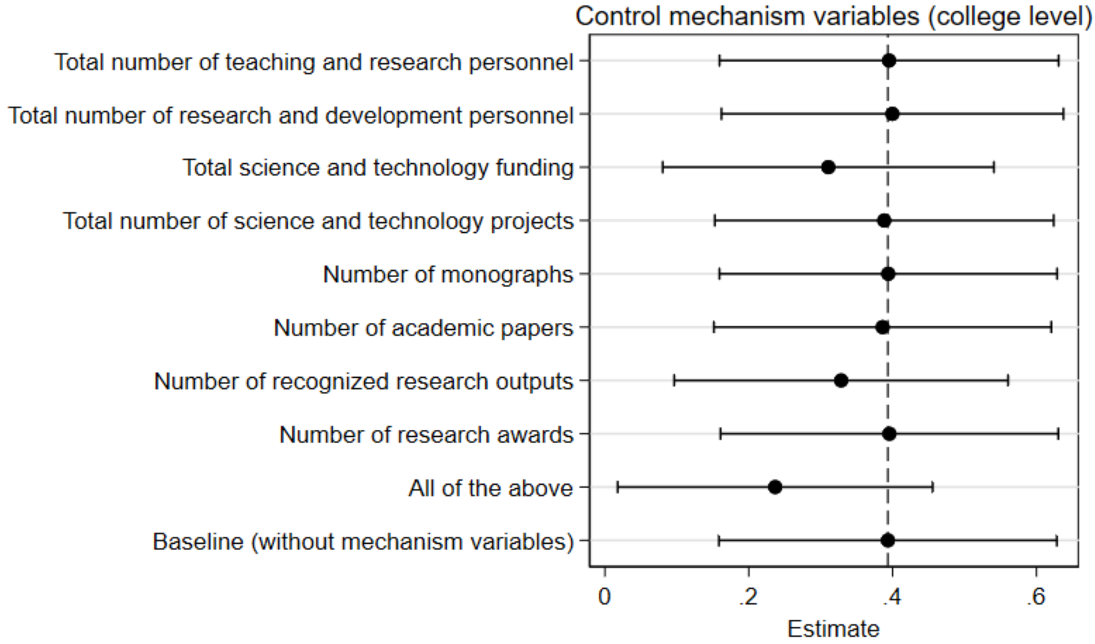
Note: Appendix Figure D3 shows the estimated coefficients and 95% confidence intervals. All mechanism variables are transformed using natural logarithms, except for trade openness. All regressions control for city fixed effects and year fixed effects. Standard errors are clustered at the city level.

The second potential mechanism through which HSR may improve college admissions quality is by enhancing college' overall strength. Colleges may benefit from the economic effects of HSR by securing increased funding. Additionally, HSR may facilitate the recruitment of more personnel and improve research efficiency. For example, Dong et al. (2020), using Chinese data on publications and citations, find that high research-output scholars tend to migrate to cities with HSR opening. Furthermore, HSR improves co-author productivity and promotes academic collaboration among researchers. It is therefore reasonable to expect that applicants prefer colleges with stronger overall strength. To examine this mechanism, we construct a set of college-level strength indicators that may attract candidates: personnel, funding, and research output. Specifically, these indicators include: total number of teaching and research personnel, total number of research and development personnel, total amount of science and technology funding, total number of science and technology projects, number of monographs, number of academic papers, number of recognized research outputs, and number of research awards. We include these potential mechanism variables as additional controls in the baseline regression to test whether HSR improves college admissions quality by enhancing college strength.

Appendix Figure D4 presents the core estimated coefficients when mechanism variables are incorporated into the baseline regression. These mechanism variables are added sequentially, and in the penultimate row, all mechanism variables are included simultaneously. The final row reports the baseline regression results without any mechanism variables, indicated by a vertical dashed line. It should be noted that since we only have *College Compilation* data from 2007 to 2017, the baseline results in the last row of Appendix Figure D4 differ from those reported in Column (4) of Table 1.

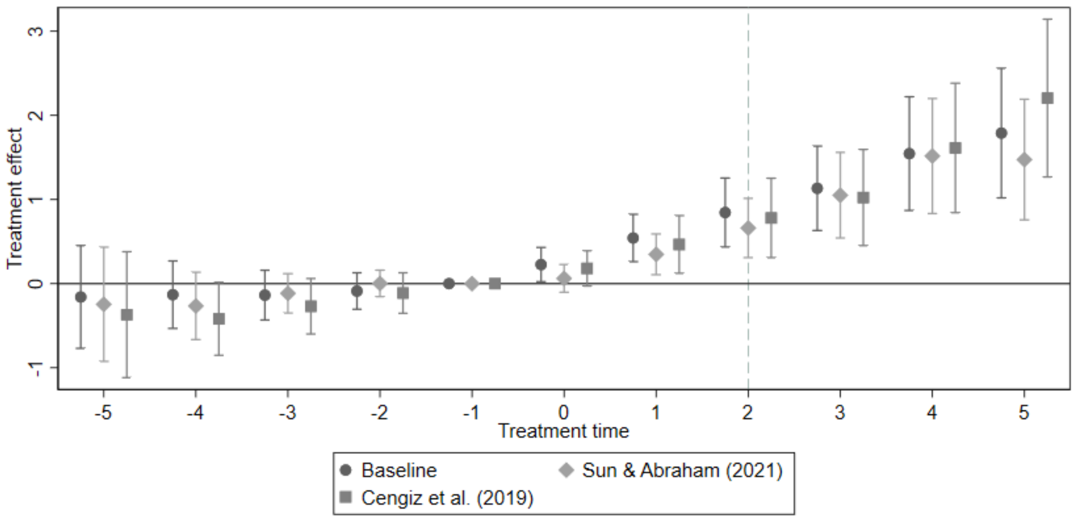
The results in Appendix Figure D4 show that, after controlling for all college-level mechanism variables, the core effect is slightly attenuated relative to the baseline regression. This attenuation may arise from the partial blocking of the causal pathway by the college strength mechanism, or it may be a mechanical effect due to the reduced sample size. First, we perform an event-study analysis similar to Figure 4, additionally controlling for all college-level mechanism variables. The results, reported in Appendix Figure D5, show that even after controlling for all college-level mechanisms, there is no evidence of significant pre-trends. Second, we regress college-level mechanism variables on HSR openings. Specifically, we consider both the contemporaneous impact of HSR openings and their lagged effects one, two, and three periods after opening. The results, presented in Appendix Figure D6, indicate that the vast majority of estimated coefficients are small and statistically insignificant.

Taken together, in the context of this study, the contribution of the mechanism whereby HSR affects college admissions quality through college strength appears to be limited. It is important to clarify that this finding does not imply that HSR has no effect on college strength, nor that candidates are unaffected by college strength. Addressing these two issues requires more in-depth and rigorous empirical analysis. This is beyond the scope of the present study and is left for future research. Moreover, we cannot control for all dimensions of college strength and have only selected representative indicators. This does not preclude the possibility that HSR enhances college admissions quality through other unobserved aspects of college strength. The correct interpretation is that the potential mechanism of HSR promoting college strength—as proxied by the selected indicators—does not explain our main finding that HSR improves college admissions quality.



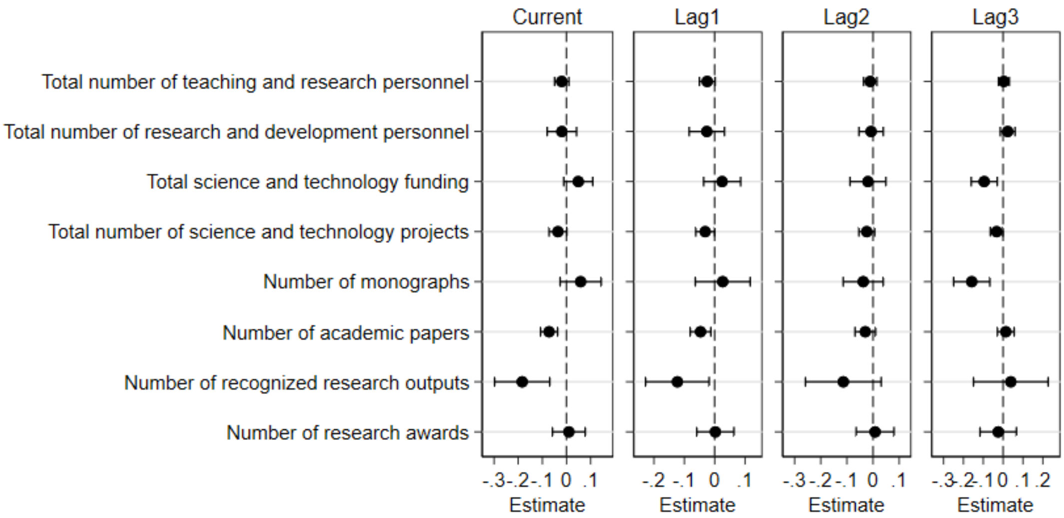
Appendix Figure D4: Potential mechanisms (HSR enhances college' overall strength)

Note: Appendix Figure D4 shows the estimated coefficients and 95% confidence intervals. All mechanism variables are transformed using natural logarithms.



Appendix Figure D5: Event study results controlling for all college-level mechanism variables

Note: Appendix Figure D5 shows the estimated coefficients β_k and 95% confidence intervals for Equation (3) and the event study results using the methods provided by Sun & Abraham (2021) and Cengiz et al. (2019).



Appendix Figure D6: The impact of HSR openings on college-level mechanism variables

Note: Appendix Figure D6 shows the estimated coefficients and 95% confidence intervals. Only the total amount of science and technology funding is log-transformed and estimated using OLS (Ordinary least squares), while all other mechanism variables are estimated using PPML (Poisson pseudo-maximum likelihood). All regressions control for college fixed effects and year fixed effects. Standard errors are clustered at the college level.

We provide evidence for the third potential mechanism by analyzing the heterogeneous effects of a city's position within the HSR network. First, we examine the heterogeneity associated with the location of HSR stations within cities. If the third mechanism—transportation convenience—is valid, we expect the effect of HSR to be stronger when the station is located in the urban core. An HSR station situated in the central area of a city enables easier access for students to reach colleges, whereas a station located in a suburban area implies higher commuting costs and weaker transportation convenience. Wang et al. (2013) found that the accessibility of HSR stations in China—such as stations located on the urban fringe—hindered the overall effectiveness of the HSR system. To demonstrate this, we plot the distribution of distances from HSR stations to city centers, as well as the distribution of distances from colleges to city centers. Using the Baidu Maps API—one of China's largest online mapping platforms—we calculate the distances from HSR stations to city centers and from colleges to city centers for the sample in this study. Appendix Figure D7 presents the density distributions of these distances. To avoid identification errors and the influence of extreme values, we exclude observations with distances exceeding 100 kilometers. We find that most HSR stations and colleges are located within 40 kilometers of the city center. Compared with HSR stations, colleges appear to be more concentrated near city centers. One possible explanation is that many HSR stations are built in suburban or county areas, farther from city centers, to serve a wider area and population.

In the Chinese administrative system, cities consist of urban districts and counties. Urban districts typically represent the core areas of a city, where most undergraduate colleges are located, while counties tend to be on the periphery. We identify the location of each HSR station accordingly: if a station is located within an urban district, we classify it as an urban HSR station; if it is located in a county, we classify it as a suburban HSR station. Based on this classification, we disaggregate the baseline HSR variable in Equation (2) into three mutually exclusive and collectively exhaustive indicators: (i) HSR stations located only in urban districts; (ii) HSR stations located only in counties; and (iii) HSR stations located in both urban districts and counties. We then replace the core independent variable in Equation (2) with these three new variables. Column (1) of Appendix Table D1 reports the regression results. We find that when HSR stations are located only in suburban areas, the effect on average admission scores is statistically insignificant. In contrast, when HSR stations are located only in urban districts, the average admission score increases by 0.287 percentage points, significant at the 10% level. Moreover, when a city has both urban and suburban HSR stations, the average admission score increases by 0.44 percentage points, significant at the 1% level.

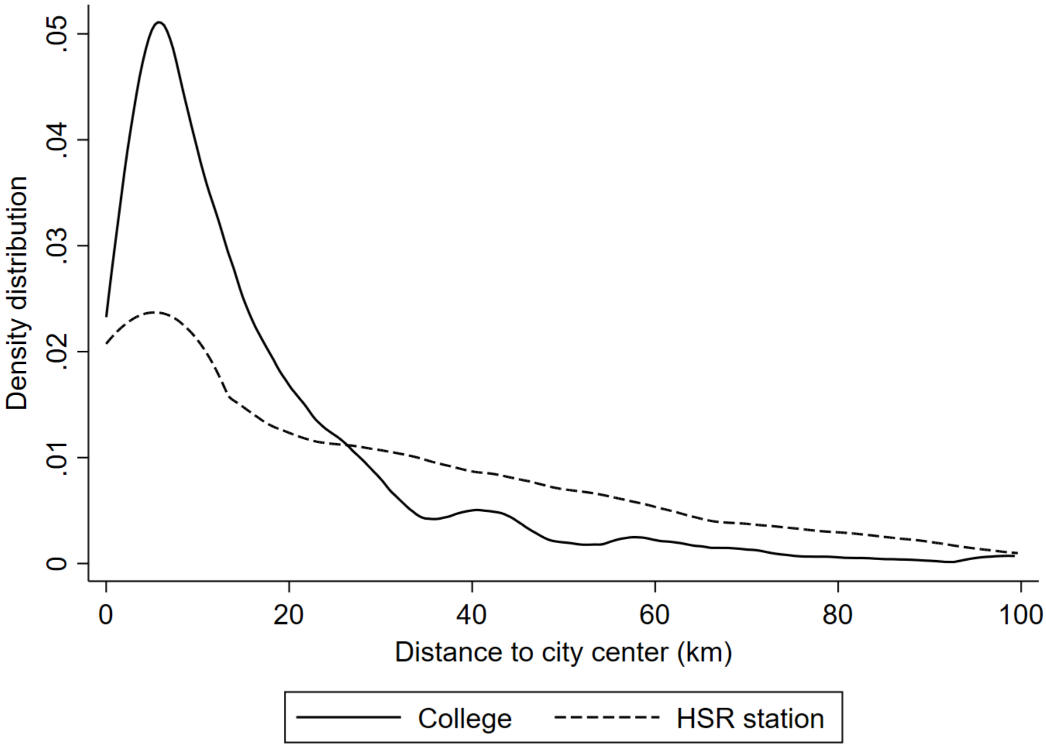
Second, we examine the heterogeneous effects based on the number of HSR lines passing through a city. If the third potential mechanism holds, we would expect the effect of HSR to be larger in cities served by multiple HSR lines. These cities are typically located at the intersections of the HSR network, offering greater frequency and diversity of train services, and thus stronger transportation convenience. To test this, we disaggregate the baseline HSR variable in Equation (2) into two mutually exclusive and collectively exhaustive indicators: (i) cities served by only one HSR line, and (ii) cities served by multiple HSR lines. We then replace the core independent variable in Equation (2) with these two new variables. Column (2) of Appendix Table D1 presents the regression results. We find that in cities with only one HSR line, HSR does not significantly improve the average admission scores of local colleges. In contrast, in cities with multiple HSR lines, the average admission score increases by 0.462 percentage points, significant at the 1% level.

Third, we exploit information on the departure and terminal stations of HSR lines. If the third potential mechanism holds, we expect stronger effects in cities that serve as the departure or terminal of HSR lines. These cities occupy critical nodes within the HSR network and benefit from greater transportation convenience. To test this, we disaggregate the baseline HSR variable in Equation (2) into four mutually exclusive and collectively exhaustive categories: (i) cities that serve only as the departure of an HSR line; (ii) cities that serve only as the terminal; (iii) cities that serve as both the departure and terminal; and (iv) cities that serve as neither. We then replace the core independent variable in Equation (2) with these four new indicators. Column (3) of Appendix Table D1 reports the regression results. We find that in cities that are neither the departure nor the terminal of any HSR line, the effect of HSR on average admission scores is statistically insignificant. In contrast, for cities that serve as key nodes in the HSR network—either as departure or terminal—the opening of HSR significantly improves average admission scores. The effect is strongest in cities that serve as both the departure and terminal of HSR lines, where the average admission score increases by 0.77 percentage points.

Appendix Table D1: The heterogeneous effects of cities' position within the HSR network

	(1)	(2)	(3)
urban district only	0.287*		
	(0.161)		
suburb only	-0.175		
	(0.239)		
Both (urban district + suburb)	0.440***		
	(0.129)		
HSR line number > 1		0.462***	
		(0.121)	
HSR line number = 1		-0.257	
		(0.328)	
HSR line departure only			0.516***
			(0.150)
HSR line terminal only			0.335*
			(0.184)
Both (departure + terminal)			0.770***
			(0.211)
All not (departure + terminal)			0.198
			(0.166)
Controls	√	√	√
College-admission province-track-batch FE	√	√	√
Year-admission province-track-batch FE	√	√	√
N	416,124	416,124	416124
R-squared	0.969	0.969	0.969

Note: The dependent variable is the average admission score aggregated at the college–admission province–track–batch–year level. Variables $urban\ district\ only$, $suburb\ only$, and $Both\ (urban\ district\ +\ suburb)$ represent: (i) HSR stations located only in urban districts; (ii) HSR stations located only in counties; and (iii) HSR stations located in both urban districts and counties, respectively. Variables $HSR\ line\ number\ >\ 1$ and $HSR\ line\ number\ =\ 1$ represent: (i) cities served by only one HSR line, and (ii) cities served by multiple HSR lines, respectively. Variables $HSR\ line\ departure\ only$, $HSR\ line\ terminal\ only$, $Both\ (departure\ +\ terminal)$, and $All\ not\ (departure\ +\ terminal)$ represent: (i) cities that serve only as the departure of an HSR line; (ii) cities that serve only as the terminal; (iii) cities that serve as both the departure and terminal; and (iv) cities that serve as neither, respectively. The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.



Appendix Figure D7: The density distributions of distances from HSR stations to city centers, as well as the distribution of distances from colleges to city centers

Note: We calculate distances using the Baidu Maps API. To avoid identification errors and the influence of extreme values, we exclude observations with distances exceeding 100 kilometers.

Appendix Table D2: The distance heterogeneity analysis results based on Table 2

	(1)	(2)
	Connecting to	
	any city in admission province	the capital of admission province
connect × ln*distance*	-0.181***	-0.546***
	(0.0340)	(0.121)
connect	0.220**	0.0367
	(0.0872)	(0.0960)
Controls	√	√
College-admission province-track-batch FE	√	√
Admission province-track-batch-year FE	√	√
College-track-batch-year FE	√	√
N	417,516	417,516
R-squared	0.976	0.976

Note: The bilateral HSR connectivity variable $connect$ is defined as the college city and the admission province being on the same HSR line. The variable $\ln distance$ represents the logarithm of the distance between the city where the college is located and the capital of the admissions province. The values in parentheses represent college city-admission province level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix E

We argue that the negative coefficient on the gap indicator does not undermine our empirical conclusions. From an econometric perspective, first, once an interaction term is introduced, the effect of the main variable is no longer determined solely by its own coefficient but also by the coefficient on the interaction term. Second, our primary focus is on the interaction between HSR opening and the gap indicator (gap × HSR). After including the interaction term, whether the estimated coefficient on the gap indicator itself can be interpreted as a credible causal effect becomes questionable.

Moreover, Appendix Table E1 employs a gap measure defined between the college province and the admission province. The results show that the estimated coefficients on the gap indicators are positive but statistically insignificant. One possible explanation is that bilateral economic development gaps may generate two opposing effects. On the one hand, higher economic development at the destination may attract students. On the other hand, higher economic development also implies larger income gap and higher living costs, which may deter students' application choices. Economic development gaps between the college province and the admission province may primarily generate an attraction effect, whereas economic development gaps between the college city and the admission province may mainly produce a deterrence effect. A thorough examination of this issue is beyond the scope of this paper and is left for future research.

Appendix Table E1: The heterogeneous effects related to regional development gap

	(1)	(2)	(3)
hsr	0.337***	0.517***	0.612***
	(0.116)	(0.123)	(0.144)
gap1×hsr	0.364*		
	(0.206)		
gap1	1.137		
	(0.958)		
gap2×hsr		0.382*	
		(0.223)	
gap2		1.164	
		(0.962)	
gap3×hsr			0.379**
			(0.184)
gap3			1.167
			(0.964)
Controls	√	√	√
College-admission province-track-batch FE	√	√	√
Year-admission province-track-batch FE	√	√	√
N	416,124	415,945	415,945
R-squared	0.969	0.969	0.969

Note: The dependent variable is the average admission score aggregated at the college–admission province–track–batch–year level. Variables $gap1$, $gap2$, and $gap3$ represent: (i) the difference in the logarithm of per capita GDP between the college province and the admission province; (ii) the difference in the logarithm of per capita GDP between the college province and the capital city of the admission province; and (iii) the difference in the logarithm of per capita GDP between the college province and the per capita GDP of the most developed city in the admission province, respectively. The values in parentheses represent city level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

To capture the outside options induced by HSR, we construct four binary HSR connectivity variables at the admissions-province level: hsr_ew for eastern provinces connected to western regions, hsr_we for western provinces connected to eastern regions, hsr_ee for eastern provinces connected to eastern regions, and hsr_ww for western provinces connected to western regions. We adopt the same definition of HSR connectivity as in Section 5, i.e., bilateral connectivity along the same HSR line.

We implement two measurement approaches: first, whether any city in the admissions province has an HSR connectivity to other regions; second, whether the capital of the admissions province has an HSR connectivity to other regions. Using these four HSR connectivity variables, we examine the impact of HSR-induced outside options on local admissions and whether this impact differs between developed and developing regions. We estimate the econometric model as specified in Equation (E1).

$$average_{n_{ipsbt}} = \alpha + \beta_1 hsr_{ew_{p,t}} + \beta_2 hsr_{we_{p,t}} + \beta_3 hsr_{ee_{p,t}} + \beta_4 hsr_{ww_{p,t}} + \boldsymbol{\gamma}\mathbf{X}_{\boldsymbol{p,t}} + \delta_{isb} + \mu_{sbt} + \varepsilon_{ipsbt} \quad \text{(E1)}$$

Here, college i is located in the same province p as the admissions province, that is, college i is situated within admissions province p. The core independent variables are the four binary HSR connectivity indicators. We include a vector of control variables at the admissions-province level, $\mathbf{X_{p,t}}$, defined similarly to the controls in Equation (2). δ_{isb} represents college-track-batch fixed effects, and μ_{sbt} represents year-track-batch fixed effects。ε_{ipsbt} is the random error term. Robust standard errors are clustered at the admission province level. The regression results are reported in Appendix Table E2.

We find no evidence that HSR-induced outside options has a significant effect on the admission scores for local students. This conclusion holds for all four HSR connectivity variables. This suggests that HSR connectivity—both between developed and developing regions, as well as within each group—do not have a significant impact on local students' scores when applying to local colleges. In other words, this effect does not appear to differ meaningfully between admissions provinces in developed and developing regions.

Taken together, we find no evidence that the admission scores of local students decline when developing regions are connected to developed regions via HSR. It should be noted, however, that the analysis in Appendix Table E2 has two limitations. First, due to the inability to directly observe local students' decision-making, the results provide only an indirect assessment. Second, the HSR connectivity indicators measured at the provincial level are relatively coarse and may not fully capture the outside options induced by HSR connectivity at the provincial level. Therefore, we are inclined to treat the results in Appendix Table E2 as suggestive evidence.

Appendix Table E2: The impact of HSR-induced outside options on local admissions

	(1)	(2)
	any city	capital
	of admission province connecting to other regions	
hsr_ew	0.994	2.079
	(2.133)	(2.129)
hsr_we	1.948	1.917
	(1.487)	(1.461)
hsr_ee	3.336	3.712
	(2.113)	(2.470)
hsr_ww	-1.681	-1.116
	(1.222)	(1.249)
Controls	√	√
College-track-batch FE	√	√
Year-track-batch FE	√	√
N	25,252	25,252
R-squared	0.893	0.894

Note: The regression sample is restricted to local admissions, meaning that the college's province is the same as the admissions province. The HSR connectivity variables hsr_ew, hsr_we, hsr_ee and hsr_ww are defined as the admission province and other regions being on the same HSR line. The variable hsr_ew is a binary indicator for eastern provinces connected to western regions. The variable hsr_we is a binary indicator for western provinces connected to eastern regions. The variable hsr_ee is a binary indicator for eastern provinces connected to eastern regions. The variable hsr_ww is a binary indicator for western provinces connected to western regions. Column (1) measures whether any city in the admissions province has an HSR connectivity to other regions. Column (2) measures whether the capital of the admissions province has an HSR connectivity to other regions. The values in parentheses represent admission province level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Appendix Table E3: Explanation and descriptive statistics for the major variables

Variable	Explanation	N	Mean	Sd Dev	Min	Max
$\Delta hflow1$	Δ Enrolled undergraduate population (normalization)	87,912	29.51	1.16	0	100
$\Delta hflow2$	Δ Enrolled postgraduate population (normalization)	87,912	25.03	0.73	0	100
$\Delta hflow3$	Δ Enrolled undergraduate and postgraduate population (normalization)	87,912	28.95	1.17	0	100
$\Delta connect$	Δ HSR connectivity	87,912	0.03	0.16	0	1
$\Delta X1$	Δ Total population (normalization)	87,912	76.44	1.00	0	100
$\Delta X2$	Δ The educated population (normalization)	87,912	75.79	1.03	0	100

Note: The variables $hflow1$, $hflow2$, $hflow3$, $X1$, and $X2$ represent different types of population flow measures from city *i* to city *j*. The variable $connect$ is the HSR connectivity indicator between cities *i* and *j*, defined as whether cities *i* and *j* are located on the same HSR line. The operator Δ denotes the difference between 2005 and 2015.

Appendix Table E4: The core effect by directional nature (eastern region and western region)

	(1)	(2)	(3)
		Enrolled	
	Undergraduate	Postgraduate	Undergraduate + Postgraduate
$\Delta connect_{ij} \times$ *East to west*	0.0123	0.0876	0.0323
	(0.0453)	(0.0633)	(0.0465)
$\Delta connect_{ij} \times$ *West to east*	0.108*	0.389***	0.196***
	(0.0602)	(0.140)	(0.0679)
$\Delta connect_{ij} \times$ *East to east*	1.279***	0.152*	1.290***
	(0.174)	(0.0778)	(0.173)
$\Delta connect_{ij} \times$ *West to west*	1.104***	0.192***	1.127***
	(0.171)	(0.0697)	(0.174)
Controls	√	√	√
City i FE	√	√	√
City j FE	√	√	√
N	87,912	87,912	87,912
R-squared	0.107	0.064	0.113

Note: The dependent variable measures the size of the highly educated population flowing from city *i* to city *j*. The dependent variable includes only currently enrolled students. Variable $connect_{ij}$ is defined as the city *i* and city *j* being on the same HSR line. The operator Δ denotes the difference between 2005 and 2015. The "East" refers to the eastern area as defined by the classification of the National Bureau of Statistics of China. The "West" includes all areas not classified as part of the eastern region under this standard. The values in parentheses represent city-pair (city i–city j) level clustered robust standard errors. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

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